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PROGRAMMING ROBOTS WITH HIGHER-ORDER REACTIVE PROGRAMMING

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2026-2027

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ROBOTS PROGRAMMEREN MET HOGERE-ORDE REACTIEF PROGRAMMEREN

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Abstract

Reactive Programming (RP) is a declarative programming paradigm based on data streams and the propagation of change. In a nutshell, if we have two variables x and y and another variable z that sums both x and y, it should be sufficient to update either x or y and z should be updated automatically.

There are several approaches to achieve Reactive Programming. One approach is to create a dedicated language. In this case the reactivity will be native to the language. Another option is to extend a language with reactive features, like

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Chapter 1

Introduction

Modern software is often written in such a way that it is event-driven, or in other words, reactive. This means that applications reacts to an incoming event by creating subsequent events that arise as a reaction to the incoming event. Such an application is often called a reactive system. Reactive systems are often implemented with callbacks or by implementing the Observer Pattern. However, this does not come without its drawbacks as this may lead to 'inversion of control' or 'callback hell'[1], [2]. To alleviate this we introduce **Reactive Programming**. A programming paradigm that allows the user to write reactive systems in a declarative way. Traditionally the sum and the product of two elements would be written as shown in Listing 1.1.

```
int x = 5;
int y = 10;
int s = x + y;
int p = x * y;
// x changes state
x = 10;
// s requires recomputation
s = x + y;
// p requires recomputation
p = x * y;

```

Listing 1.1: Imperative product and sum

Whenever x or y would change state, you would have to manually update s and p. Reactive Programming allows you to declare x and y as sources of data and s and p as the result of the relevant computations. Whenever the value of x or y would change, the variables dependent on the sources would be automatically recomputed, and the result propagated further in the reactive system. Schematically the dependency graph of product-sum would look like Figure 1.1 There are different ways of how such a reactive system could be created using Reactive Programming. One way is to use a programming which itself is natively reactive. A few examples are Haai [3], Another is to add native reactivity to an existing language by

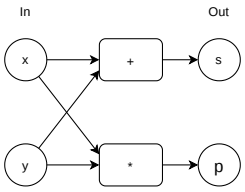


Figure 1.1: Dependency graph for product-sum

extending it the language or by implementing a reactive library such as ReactiveX. We will be looking at the former and more specifically the higher-order reactive programming language **Haai**, which was developed at the SOFT lab at the VUB. Haai uses reactors as their As a preview, an example of how such a product and sum could be written in Haai is shown in Listing 1.2. Haai uses reactors as its native way of declaring a reactive program.

```
(defr (product-sum x y)
    (def sum (+ x y))
    (def product (* x y))
    (out product sum))

```

Listing 1.2: Product sum in Haai

Additionally, we want to delve into the world of electronics. From maintaining your fridge's temperature to controlling a drones' altitude, electronics are deeply embedded in everyone's day-to-day life. Writing software for these electronics can also be challenging. This mainly due to the fact that using peripherals require using their associated libraries, which are not always standardized. Another challenge is that creating programs for microcontrollers often require the developer to use a low-level programming language such as C/C++ or Rust, which come with their own challenges. Nowadays, there are some efforts being made to provide a standardized collection of tools to aid with the development of systems with many peripherals, or robots. There is Robot Operating Systems (ROS) [4], Open Robot Control Software (OROCOS) [5], Yet Another Robot Platform (YARP) [6] to name a few. We will be focussing on using **(ROS)**.

1.1 Contributions

Reactive Programming is used across a multitude of domains. It is used in web and mobile development for instance. However, we will be focussing on using it for the Internet of Things (IoT). IoT consists of multiple physical objects, usually with very limited computing power and/or memory. In this thesis we introduce a way of declaring a reactive program across multiple devices from a single source. This means that the developer declares the program once, along with declarations where which part of the program lives and then deploys the program to the relevant devices making use of ROS. This is achieved by the following components:

- An extension of the Remus VM [7], more specifically, extending the language to accept a declaration of where which part of the program lives. The Remus compiler will use this metadata to split the Haai reactor declarations into an embedded part and a server part. Each piece will then be available for the user to be deployed in their respective parts of the IoT network.
- A new tool called CHaai, which translates the Remus bytecode obtained from compiling the Haai program into a C program. The C program will be required to run the program on microcontrollers using ROS.
- A C port of the existing Remus VM written in rust [8], also created to allow developers to create reactive programs for microcontrollers. However, it is adapted with respects to the current Instruction Set Architecture (ISA) of the Racket version of Remus [7] and allows usage of ROS.

1.2 Overview of this Dissertation

Chapter 2 gives the problem statement as well as an overview of the current Reactive Programming solutions and existing toolkits to program networks of microcontrollers. It also introduces a concept called 'Tierless programming', which will inspire the declarative way of telling Remus where which part of the reactive program lives.

Chapter 3 introduces μ-Remus. A C port of the Remus VM, which was previously written in Rust as well as in Racket. It explains why the C port was necessary as well as conceptualizes the link between the ROS communication protocols and the dependency graph generated by Reactive Programming.

Chapter 4 demonstrates usage of the Haai language extensions inspired by Tierless programming to declare where which part of the reactive program lives. It will also tell you about the CHaai tool. Which will create a C version of the reactive program. It is ported to C to be compatible with ROS.

Chapter 5 will evaluate the tools and extensions driven by an example. The target microcontroller will be a Raspberry Pi Pico H (2021) and will be used along a PC.

Lastly, chapter 6 will conclude this dissertation and give notes on some possible future work.

1.3 Approach

We first start by discussing the state of the art, what tools already exist and then move on to the problem statement and continue with the contributions part of the dissertation. However, since it is heavily dependent on a language and a toolkit we will first give a brief introduction into some relevant parts of the language or toolkit, each time providing the necessary context and information, prior to moving on to the actual contributions.

Chapter 2

State of the Art

Many programs spend most of their time reacting to events or their environment: a thermostat displays the temperature a sensor picks up, a lightbulb reacts to the pressing of a button, a notification is pushed once an email arrives. Reactive Programming (RP) is a programming paradigm that suits these kinds of programs. Instead of writing code that explicitly updates values when something changes, the programmer declares how values depend on one another and the language takes care of propagating changes through those dependencies [1].

Embedded devices are prime examples of devices that spend their lifecycle reacting to their surroundings, often taking part in a much larger system. Consider, for instance, a modern-day car, which is equipped with a myriad of subsystems. A braking system reacts to an obstacle drawing too close to the front of the car while a dashboard behind the steering wheel reacts to the metrics fed by other subsystems such as temperature, speed, gear shift, etc.... All these subsystems communicate with each other and depend on one another to form a much larger, coherent system. Inherently, these systems are distributed as they span multiple devices. Robots are no different: their sensors, actuators and controllers often run on separate boards that must also cooperate.

Programming such a system is not trivial. Each device runs its own program, written for its hardware. Additionally, the developer has to connect these devices through a medium that allows them to communicate with each other. While it is possible to write reactive programs for all these devices and then glue them manually, doing so is cumbersome [1], [9]. Creating a single RP program and avoiding this manual glue is the challenge of this dissertation. Moreover, we want to build on an existing robotics platform, so that its software tools and libraries can aid the development of future robotics software. The motivation behind choosing RP as our paradigm is given together with the necessary background in Section 2.1.

The following requirements arise. First, we need to choose an RP language that runs on microcontrollers (MCUs). This is discussed in Section 2.2. Second, we need a robotics platform that has an SDK and a user base, and runs on MCUs. Preferably, it also abstracts away the communication between devices, independent of the medium used (e.g. UART, Bluetooth). This is discussed in Section 2.3. Third, the robotics platform has to work together with the RP

language. This is discussed in Section 2.4.

2.1 Reactive Programming

Software nowadays is largely event-driven, meaning that programs spend most of their time reacting to incoming data. In imperative languages this is traditionally implemented with handlers, callbacks or the Observer pattern. The programmer registers a function with an event source and the runtime calls that function when that event occurs [1].

Unfortunately, this scales poorly as it gives rise to two problems: callback hell and inversion of control [1]. Callback hell occurs when callbacks are so deeply nested that it becomes difficult to tell what each callback does and when it is invoked. An example is shown in Listing 2.1, where a temperature sensor reads its pin, normalizes the read voltage into temperature and displays it on a screen. Any errors while displaying or sending are logged.

```
readSensor((voltage) => {
    normalize(voltage, (temperature) => {
        display(temperature, (errors) => {
            log(errors, () => {
                sendToServer(temperature, (errors) => {
                    log(errors, () => {
                        // etc...
                    })
                })
            })
        })
    })
})
```

Listing 2.1: Illustration of callback hell in JavaScript

Inversion of control is related to callbacks: instead of the programmer's code being in charge, the programmer hands the function to a framework or event source and the framework decides when it is called. As a result, the control flow is no longer visible in the program text as the framework decides how and when the programmer's code is run. In Listing 2.2 we define two functions `hello` and `foo` within the Flask web framework. These functions are registered to be called when their respective endpoints `"/hello"` and `"/foo"` are requested. The programmer never calls them directly. Additionally, the flow between receiving the request and the registered function being called is hidden inside the framework.

Reactive Programming (RP) attempts to solve these problems [1]. RP is a declarative programming paradigm that allows the developer to declare a program in terms of dependencies, instead of having to manually manage these dependencies [1]. A value that is computed from other values is said to be dependent on them. When these values in turn also have other values that they depend on, they form a dependency chain. In an imperative language, you would have to manually update all these values in the correct order. In RP, the

```python
@app.route("/hello")
def hello():
    return "world"

@app.route("/foo")
def foo():
    return "bar"
```

Listing 2.2: Inversion of control in Python, using the Flask web framework

actual propagation of the state change in the dependency chain is abstracted away in the language design. For instance, in a program you could have two variables `x` and `y` that hold values and variable `s` which holds the sum of `x` and `y`. Under the imperative paradigm, whenever `x` or `y` would change state, it would require manual recomputation of `s` as shown in Listing 2.3.

```c
int x = 5;
int y = 10;
int s = x + y;
printf("%d", s); // 15
x = 10; // state change
printf("%d", s); // 15
// requires manual recomputation of s
s = x + y;
printf("%d", s); // 20
```

Listing 2.3: Manual recomputation of `s` in C

In Reactive Programming, it is sufficient to update either `x` or `y` and the change is propagated throughout the dependency chain. A dependency chain can also be seen as a directed acyclic graph (DAG), with input variables and output variables, often called *signals* [1]. Converting Listing 2.3 into a DAG yields the graph shown in Figure 2.1. In Listing 2.3, the programmer has to maintain these dependencies manually; in RP, the language does so automatically. In essence, whenever an input signal changes state, the change is propagated through the graph and finally updates its output signals. An example in the reactive language REScala is shown in Listing 2.4. The input signals `x` and `y` are declared using the `Var` class and the dependent signals can be derived with `Signal { ...}`, as seen in the example with `sum`. In this example we can see that `s` is declared once through the `Signal` dependency, instead of nested callbacks. Whilst RP does not fully solve inversion of control, since the runtime still decides when it is called, it does make the data flow explicit [10].

2.1.1 Hosted and independent reactive languages

In practice there are two kinds of reactive languages. The first kind are *hosted* reactive languages, which are built on top of an existing host language. Examples are REScala, built on

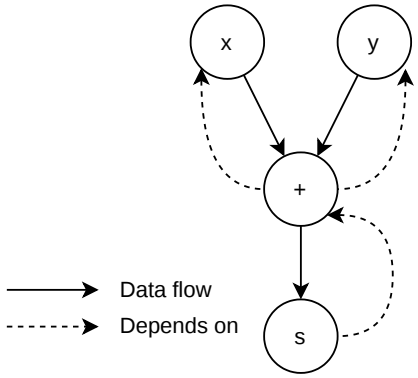


Figure 2.1: Dependency graph for `x`, `y` and `s`

```scala
def sum(a: Signal[Int], b: Signal[Int]): Signal[Int] = Signal { a.value + b.value}

@main def demo(): Unit =
    val x = Var(5)
    val y = Var(10)
    val s = sum(x, y)

    println(s"s = ${s.now}") // s = 15
    x.set(10)
    println(s"s = ${s.now}") // s = 20
```

Listing 2.4: Automatic recomputation of `s` in REScala

top of Scala, and Yampa, built on top of Haskell. These RP languages inherit the tools of their host language such as compilers, runtime, and libraries [1]. These reactive languages consist of two parts: a reactive part and a non-reactive part of the code. The reactive part is the part that the host language is extended with. The non-reactive part is functions or code of the host language that work on signals through a mechanism called *lifting*. However, lifting has two drawbacks. First, it can lead to *reactive thread hijacking*: the runtime has to wait for the lifted function to return, so a function that never returns takes over the runtime and makes the program unresponsive. Second, lifting can cause a *reactive/imperative impedance mismatch*: the runtime, not the programmer, decides the order in which the lifted functions run, so side effects such as I/O may happen in an unintended order and lead to faulty behaviour [3].

The second kind are *independent* reactive languages. These languages were built from the ground up with their own compilers, runtime, tooling, etc. Some examples are Haai, EmFRP, CFRP and Elm [1]. This does not mean that they avoid lifting, since some of these languages still use a non-reactive part to feature data structures and functions within the reactive part [1]. An independent reactive language lacks the ecosystem of a host language, and reusing existing code requires a foreign function interface. In return, its designers control the entire implementation, so they can enforce the properties their target domain needs, such as safety and efficiency [1].

An extreme case is *Haai*, a pure reactive programming language, which has no non-reactive parts [3]. In Haai a reactor is the only computational unit, which is essentially a description of the DAG shown in Figure 2.1. Instantiating a reactor is called a *deployment*, which supplies the reactor with its concrete input signals, called *sources*, and creates its output signals, called *sinks* [3]. The benefit of Haai is that everything is a reactor; even primitives such as `+` and `-` are reactors, and not functions. This means that there is no lifting and consequently Haai does not suffer from the drawbacks of lifting (thread hijacking and impedance mismatch) [3]. An example of a Haai program can be seen in Listing 2.5. `defr` is used to declare a reactor along with its name and its sources `x` and `y`. `def` is used to define internal signals and `out` is used to declare one or multiple sinks, though if there is only one sink, ending the reactor with it is sufficient instead of using `out` [3]. In Listing 2.5 we define `s` as an internal signal, which is the result of the deployment of the `+` reactor. The signal `time` is a built-in signal that increases its value over time. Should the values of the sources `x` or `y` change, the change is automatically propagated to the deployment of the `+` reactor and to the sink `s`. In the case of `total`, which is the result of the deployment of the `sum` reactor with the sources `time` and `10`, the sum is recomputed each time the value of the `time` signal changes.

```
(defr (sum x y)
    (def s (+ x y))
    (out s))

(def total (sum time 10))
;; total = 10, 11, 12, ...
```

Listing 2.5: Reactive program in Haai

2.2 RP languages targeting MCUs

Reactive Programming (RP) has many uses, from web applications that react to the input of users to microcontrollers (MCUs), which react to signals from hardware [1], [11]. In this thesis we focus on RP for MCUs, which are used in a growing number of domains, such as the Internet of Things (IoT) [12], edge computing [13] and artificial intelligence (AI) [14].

MCUs have little memory and processing power, and support few programming languages, mostly low-level ones such as C, C++ and Rust [15]. This is also why it is hard to use hosted reactive languages on these devices: they often rely on the runtime of their host language [1], [7]. For instance, Yampa, a language built on top of Haskell, cannot be used on most MCUs because the Haskell runtime is too large to be flashed onto them [16]. Despite this, there have been efforts to get reactive languages running on MCUs.

To compare these languages, we use the following requirements:

- *Predictable memory footprint*: the program's memory usage must be known in advance to ensure it meets the memory constraints of the MCU.

- *Access to sensors and actuators*: the program must be able to read sensors and drive actuators, either through the language itself or through MCU-compatible code.
- *Interoperability*: the reactive language must be able to interoperate with a low-level language that the robotics platform is written in.
- *Portability*: the reactive program must be able to run on different MCUs within a system, or robot.

Table 2.1 gives a high-level overview of independent RP languages that were designed to run on MCUs, and whether they meet the requirements above.

Some honourable mentions that fell just short of Table 2.1 are Hailstorm, ReactiFi and RT-FRP. Hailstorm runs on top of the Erlang virtual machine, which occupies about 30 MB of memory [17]. Like the Haskell runtime, this exceeds the memory of most MCUs. Additionally, running Hailstorm on MCUs was left as future work, and in the presence of recursion its memory usage cannot be predicted statically [17]. ReactiFi specifically targets Wi-Fi chips [8], [18], so unlike the other languages in Table 2.1, it is not a general-purpose language for MCUs. RT-FRP statically bounds the time and memory of each execution step [19], but its focus is on the formal integration with a base language rather than on implementing the language. Its successor E-FRP, which is included in the table, shifts the focus to implementation and was compiled to run on an MCU [20].

Language	Predictable memory	Sensors & actuators	Interoperability	Portability
EmFRP [16]	✓	✓	✓	∼
CFRP [21]	∼	✓	✓	∼
Haai + Remus [7], [8]	✓	✓	×	✓
Juniper [22]	×	✓	✓	∼
E-FRP [20]	✓	✓	✓	∼

Table 2.1: Candidate RP languages for MCUs against the requirements. ✓: met, ∼: partially met, ×: not met or not shown in the cited work.

As Table 2.1 shows, none of these languages meets all the requirements. However, some requirements, such as interoperability, can be met by building additional tooling. Juniper has no predictable memory footprint [22], so we leave it out. E-FRP has a similar design to EmFRP [11], but it is a prototype that generates C for the compiler of one specific MCU, the PIC16C66 [20]. EmFRP, in contrast, generates platform-independent ANSI C that can run on multiple MCUs, and offers more language features, such as modules, algebraic data types and type inference [16]. Therefore, we compare the following languages: EmFRP, CFRP and Haai + Remus.

2.2.1 EmFRP

EmFRP is a reactive language designed to work on small-scale embedded systems that cannot run a full-fledged operating system such as Linux [16]. EmFRP works as follows: the developer declares a series of nodes, which can be seen as signals. These nodes can refer to other nodes,

which defines the dependencies between nodes. These connected nodes form a directed graph. EmFRP does not allow recursion; instead, EmFRP allows cycles through the `@last` annotation, which allows referring to the previous value of any node [16].

EmFRP's prototypical example is a thermostat, shown in Listing 2.6. We start by declaring the module name, the program's inputs, in this case the sensed temperature and the periodical pulse, and the outputs, in this case the heater state (on/off), followed by the used libraries. Afterwards, a single node is declared that periodically turns the heater on if the temperature is below 50 and off otherwise [16]. Between pulses, the heater keeps its state.

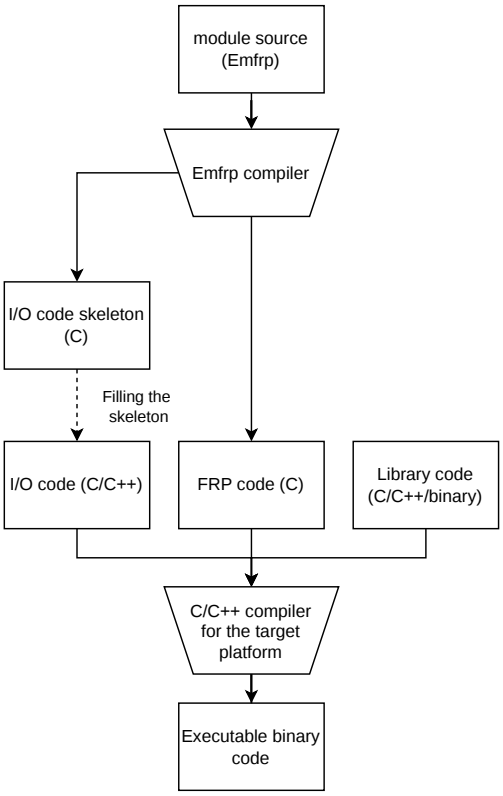


Figure 2.2: EmFRP development workflow [16]

2.2.2 CFRP

2.2.3 Haai + Remus

An example would be **EmFRP**, an FRP language created by K. Sawada and T. Watanabe that mainly targets small-scale embedded systems [16]. Like Haai, EmFRP is a graph-based RP language. This means that EmFRP follows the same philosophy in terms of values that depend on other signals and automatic change propagation. More concretely, a EmFRP program is written in terms of nodes, who may refer to other nodes that it depends on. Conceptually it works the same as the DAG that Haai generated in Figure 2.1.

However, since EmFRP is designed to work on small-scale embedded systems, some design considerations have been made. For instance, the dependency graph must be known statically, meaning it cannot change during runtime. Additionally, it does not allow recursion in function or in type definitions and nodes must always have a name. This because they are not first class values in the language.

```
module Thermostat
in
    temperature : Int,  # temp sensor
    pulse1min : Bool    # periodical pulse
out
    switching : Bool    # heater state
use
    Std                 # standard library

node init [ False ] switching =
    if pulse1min then
        temperature < 50
    else
        switching@last  # previous value
```

Listing 2.6: Thermostat controller in EmFRP [16]

```
#include "Thermostat.h"
void Input(int *temperature, int *pulse1min){
    /* Your code goes here ... */
}

void Output(int *switching){
    /* Your code goes here ... */
}

int main(){
    ActivateThermostat();
}
```

Listing 2.7: I/O Code Skeleton for Thermostat [16]

Furthermore, interacting with hardware requires using magic constants such as memory addresses, that are architecture- or peripheral specific. To address this, the EmFRP compiler generates an I/O code skeleton to connect the program with the hardware. The developer remains responsible for completing the skeleton code though.

The way EmFRP works is as follows: the developer creates a series of EmFRP modules, the EmFRP compiler then generates I/O skeleton code in C and FRP code in C. The developer is then required to fill in the I/O skeleton code. A schematic of the workflow can be seen in Figure 2.2.

In the paper introducing EmFRP, they show an example with a thermostat, where each minute, the temperature is read and depending on the temperature value, the heater is turned on or off. The code for this example can be seen in Listing 2.6. The EmFRP program is then compiled into the I/O skeleton shown in Listing 2.7 as well as the FRP code in C. Once the skeleton code is completed, the program can be further compiled by the relevant C compilers into the required binaries.

Another RP language targeting embedded devices is CFRP [21]. By the same creators of EmFRP, K. Sawada, T. Watanabe, as well as K.Suzuki and K. Nagayama, CFRP was created as a pure FRP language that compiles to stand-alone C++ code. It was created to show that FRP is beneficial for developing software for small-scale embedded devices.

```
1  -- Including C++ header files
2  %{
3  #include "sensors.h"
4  #include "ac_ctrl.h"
5  %}
6
7  -- Declaring identifiers connected to external devices
8  %input tmp :: Signal Float = sen_tmp;
9  %input hmd :: Signal Float = sen_hmd;
10 %import ac :: Signal Bool → Signal () = ctrl_ac;
11
12 -- Main expression
13 let di t h = t - 0.55 * (1 - 0.01 * h) * (t - 14.5) in
14 ac (lift1 (\x → x > 24) (lift2 di tmp hmd))
```

Listing 2.8: Example program in CFRP [21]

It was designed as follows: A CFRP program consists of a series of declarations, which denote signals and links to hardware (in C++). Followed by a single expression that builds to reactive system. An example can be seen in Listing 2.8 where the air conditioner is turned on or off depending on a discomfort index. The discomfort index is calculated from the temperature (tmp) and humidity (hmd). Notably it seems to use the same 'base' datatypes that C++ provides but signals get an additional annotation (the Signal type). This is to make the datatype predictable at compile time.

Furthermore, it uses a single Haskell-like expression to construct a dependency graph, just like EmFRP and Haai do. Which is then used to compile to the C++ program. The caveat with CFRP is that C++ objects and functions referred to in the CFRP file have to be declared in

separate C++ files. This means that only the 'global' reactive system is declared in CFRP and the rest in C++.

Whilst there are more RP languages targeting embedded systems like Atom [23] or Copilot [24]. They seem all to compile to a low-level programming language such as C, C++, Rust or Zig. Though the more customary languages that these RP languages compile to are either C or C++.

At the VUB a virtual machine (VM) was created for Haai to be able to run Haai on the bare metal. The VM is called Remus [7], which was later re-implemented in Rust and with some modifications a prototype was created to run a Haai program on a M5Stick [8]. The benefit of maintaining a scheme/racket-like syntax with dynamic typing remains, at the cost of having to compile the Haai program to Remus bytecode. Which can then be interpreted by the Rust based implementation of Remus.

At this point one could ask him or herself what the point is, of examining these RP languages that target MCUs. Well as mentioned in Section 2.2, due to the increase in use of microcontrollers. Though in this thesis we want to focus more on a system of these embedded devices. Take a robot for example: A robot has sensors as input and has other peripherals that are then used to interact with the real world, depending on these inputs. In other words, the robot reacts to the sensors by instructing the motors what to do or where to move next.

To build this system of embedded devices we could create some reactive programs in an RP language of choice that targets these devices, but this would leave the developer responsible for implementing all the communication between the devices as well as some central computing unit that coordinates all these devices and peripherals. Most of the time requiring the developer to use a library that uses some kind of communication protocol under the hood (Bluetooth, UART, I2C, SPI, USB, etc...). Easier would be if we could use some kind of standardized toolkit that saves the developer from these complexities, which is what we will be looking into next.

2.3 Systems to program robots

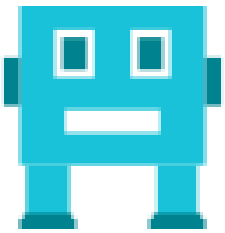
In this section we are going to have a look at the following systems that aid development in robotics: Open Robots Control Software (OROCOS) [5], Robot Operating System (ROS) [4], [25], [26], [27] and Yet Another Robot Platform (YARP) [6].



(a) OROCOS logo



(b) ROS logo



(c) YARP logo

2.3.1 OROCOS

OROCOS, started at the Katholieke Universiteit Leuven (KUL), as a project aimed at becoming a general-purpose and open robot control software package. On a more technical side of things, it is software for all sorts of robotic devices and computer platforms (operating systems). OROCOS features software components for kinematics, dynamics, planning, sensing, control, interfacing, etc... through a middleware. The system also provides features in the form of modules that can be split up in the following categories:

- Supporting modules

  Software that does not interact with the hardware directly but rather supports the interaction with it. These modules can range from simple mathematical modules to aid vector operations, configuration modules, simulation modules to real-time operating system modules.

- Robotics modules These are the modules specifically interacting with the hardware such as kinematics and dynamics modules or servo controller modules. These robotics modules tend to use one or more of the supporting modules to make the usage or development of these robotics modules easier.

- Components These are the Common Object Request Broker Architecture (CORBA) with their Interface Definition Language (IDL) descriptions. These components are built upon the previous modules and are used as the building blocks of robots, as they allow for multi-machine inter-process communication.

Unfortunately OROCOS does not allow it to be used on the bare metal. Meaning it is only usable on a system that has an OS. However, with the first version of ROS, the real-time toolchain (RTT) of OROCOS, called OROCOS RTT was often used alongside ROS. Nowadays, the second version of ROS, or ROS 2 is used in favour of ROS + OROCOS RTT, or the OROCOS Component Library (OCL) as ROS 2 absorbed both the toolchain and component features of OROCOS and thus no longer needs them as middleware. The hardware libraries for Kinematics Dynamics (KDL) or Bayesian Filtering (BFL) are still used. OROCOS also provides support to interoperate with other robotics systems such as ROS and YARP. To summarize, OROCOS seems inadequate as a robotics system to build a distributed system of embedded devices because it does not run on the bare metal. Additionally, it is not used as a communication layer across devices and actually depends on other robotics systems to bridge the communication such as the next system. OROCOS could however, be used as a supporting tool, depending on the domain of robotics.

2.3.2 ROS

Originating from Stanford, Robot Operating Systems, or ROS, is in fact not an operating system but rather a Software Development Kit (SDK). Like OROCOS, ROS provides modules to aid in the development of robotics systems. The modules can be categorized as follows:

- Middleware: This encompasses communication among components, from network APIs to message parsers.
- Algorithms: Commonly used algorithms when building robotics applications
- Developer tools: A suit of CLI tools for configuration, launch, introspection, visualization, debugging, simulation and logging to name a few.

The middleware is essentially a structured communication layer above the operating systems of a compute cluster [25]. A ROS cluster consists of several on-board devices and off-board devices, which can connect and disconnect at runtime. Consequently, the designers of ROS chose for a peer-to-peer (p2p) system as having a central server would mean a lot more network saturation as messages would have to a central point and would then have to be forwarded to their respective recipients. In a p2p system each message would be forwarded to an intermediary recipient who has the address of the final recipient or another intermediary, leading to less network saturation and shorter distances travelled.

The philosophy of ROS of using multiple standalone components within its system allows for developers to reuse or repurpose existing hardware libraries within ROS, but without actually depending on ROS. This also allows for robust testing of these libraries, without being dependent on ROS. The decoupling of these components also means that they can be used in other robotics systems where there is no inherent dependency and vice versa, giving ROS access to existing standalone components, such as drivers and vision algorithms.

As previously mentioned, the designers of ROS chose for a p2p system. However, the actual implementation is divided into nodes, messages, topics and services. Nodes are processes that perform computations. This means that each device in the system could consist of one or more nodes. Since these nodes also communicate with each other, we could create a graph with the edges denoting where the nodes send their messages to.

Figure 2.4 shows a robotics example where some products are being transported down a conveyor belt. However, these products have to be classified by colour, which the sensors on the 'sensors' device can detect. The sensors can then send the detected colour to the operator computer, which selects the appropriate sorter to sort the product. Each sorter consists of an arm and a sensor. The sensor detects when the product is in the sorting range and sends a signal to the arm, which sorts it. Furthermore, the operator computer shows the detected colours on the screen and allows the user to manually operate a sorter if need be thus completing the graph shown in the figure.

```
1 fieldtype1 fieldname1
2 fieldtype2 fieldname2
3 fieldtype3 fieldname3
```

Listing 2.9: IDL example (pseudocode)

Messages that these nodes send to each other are actually strictly typed data structures. These can consist of the primitive types (integer, floating point, boolean, etc...), arrays of primitive types as well as constants though they must be defined in a `.msg` file, or an interface definition language (IDL). An example can be seen in Listing 2.9. Messages can also be composed of

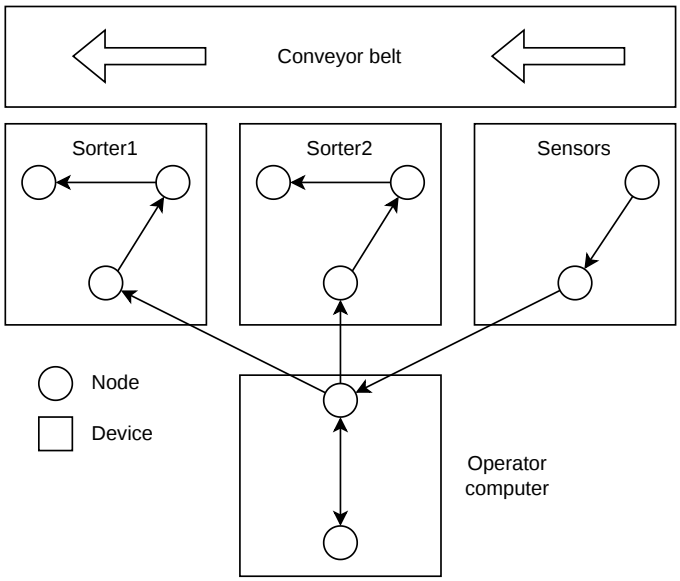


Figure 2.4: ROS node graph example

other messages or arrays of messages which can be nested arbitrarily deep. Do note that the types used in the IDL depend on the language where the ROS library is used. If it is used in C++ you can use the native C++ primitive types. In Python, you would have to use the library-provided types that map to their respective C++ types. The ROS Client library that provides the API to create a ROS graph is called rcl, which is written in pure C. Depending on the language used, the API is built upon further resulting in rclcpp (C++), or rclpy (Python).

To be able to broadcast messages asynchronously to multiple devices, ROS sends messages through a topic. Nodes subscribe to a particular topic if they wish to receive messages published to the topic. This allows for a many-to-many relationship within the ROS node graph. A re-illustration of Figure 2.4 can be seen in Figure 2.5, this time illustrated with the topics, where unidirectional one-to-one, one-to-many, many-to-many or many-to-one relationships are allowed. A bidirectional relationship between two nodes can be created by making one of the nodes a subscriber to a topic and the other a publisher as well as the inverse on another topic as seen on the operator computer.

Interestingly, we can use topics to create a graph-like structure, just like the dependency graphs that are generated from an RP program in Haai, EmFRP or CFRP. Schematically we can recreate the DAG defined in Figure 1.1 in a ROS system, as shown in Figure 2.6.

Whilst topics are the main communication protocol that ROS uses, it is lacking for synchronous communication or short-lived communication that expects a response from a request. To solve this, the designers implemented services, which works analogous to the web services. A node places a request to a uniquely named node which returns a response after processing the request, just like web services. Like topics, service definitions must be stored in a `.srv` file.

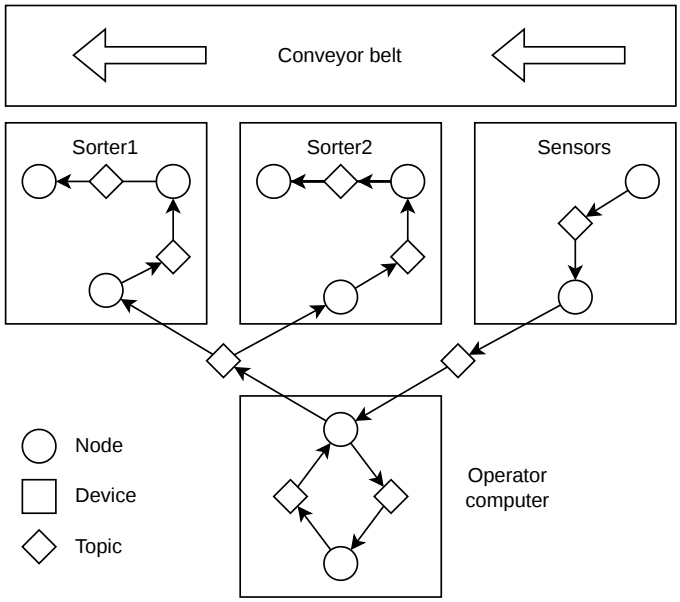


Figure 2.5: ROS node graph with topics

In case neither topics nor services are inadequate and, you need the best of both worlds there is also actions. Which is a combo of both topic and requests and ideal for long-running tasks and works as follows: it starts off like a service, a node sends a request to a uniquely named node, which in turn has to process the request. During the processing of the request the uniquely named node produces some feedback which gets published to a topic and is sent to its subscribers. After processing the request a response is sent back to the node that made the request. Unlike services or topics, actions allow for cancellation of the request if needed. Like services and topics, actions must be defined in an `.action` file.

At the end of the previous section we mentioned ROS 1 and ROS 2. Essentially ROS 1 is the predecessor of ROS 2 as the number probably already suggested, however they are vastly different. As ROS 1 grew in popularity across robotics, its limitations became increasingly evident. Security, reliability in nontraditional environments, and support for large-scale embedded systems became essential [26]. Since ROS 1 did not provide solutions for these concerns, companies were forced to create workarounds. To solve these issues they rebuilt ROS from the ground up and thus ROS 2 was born. A summary of the comparison of features between the versions can be seen in Table 2.2. Without getting in to too much of the technical changes, the most noteworthy changes would be the new underlying network communication, Data Distribution Service, which is also used in critical infrastructure. The peer-to-peer discovery and the support for embedded systems.

The change to DDS as its network communication solved much of the problems as DDS has best-in-class security, embedded and real-time support, multi-robot communication, and operations in non-ideal networking environments [26]. Peer-to-peer discovery is also one of its

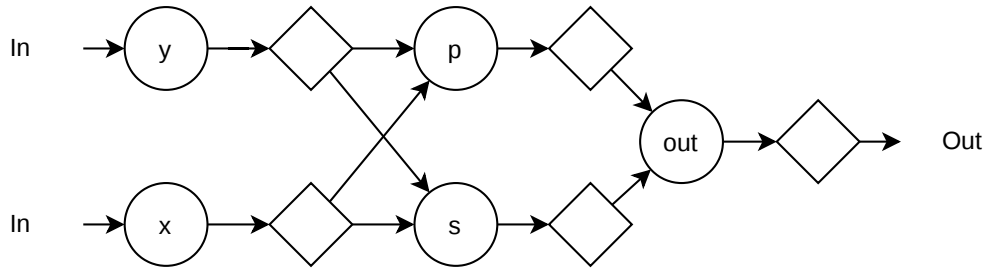


Figure 2.6: Product sum in a ROS node system

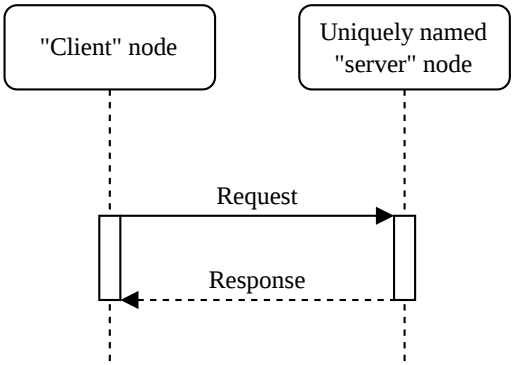


Figure 2.7: ROS service protocol

core defining features, as previously discussed.

Micro-ROS

ROS now also supports embedded systems. It does so because a robot is assumed to have sensors, actuators or other peripherals. These robots could contain microcontrollers that need to communicate with CPU(s) running ROS 2. ROS 2 supports embedded systems by providing a dedicated SDK called micro-ROS [27] for resource constrained devices (such as microcontrollers). The SDK provides all major features that the ROS 2 SDK provides by reusing as many packages as possible and providing them in a C by extending the existing rcl libraries with the rclc packages.

To be able to provide all these major features, micro-ROS relies on an open source implementation of DDS for eXtremely Resource Constrained Environments [28] (XRCE or XRCE-DDS) by eProsima called micro XRCE-DDS [29]. Micro XRCE-DDS comes with an agent that bridges the gap between XRCE-DDS and standard DDS, allowing microcontrollers to communicate with CPU(s) running ROS 2 seamlessly, independent of the underlying communication mediums (UART, I2C, Bluetooth, Serial, etc. . .).

Unfortunately, micro-ROS cannot be used on every single microcontroller. As ROS 2 was

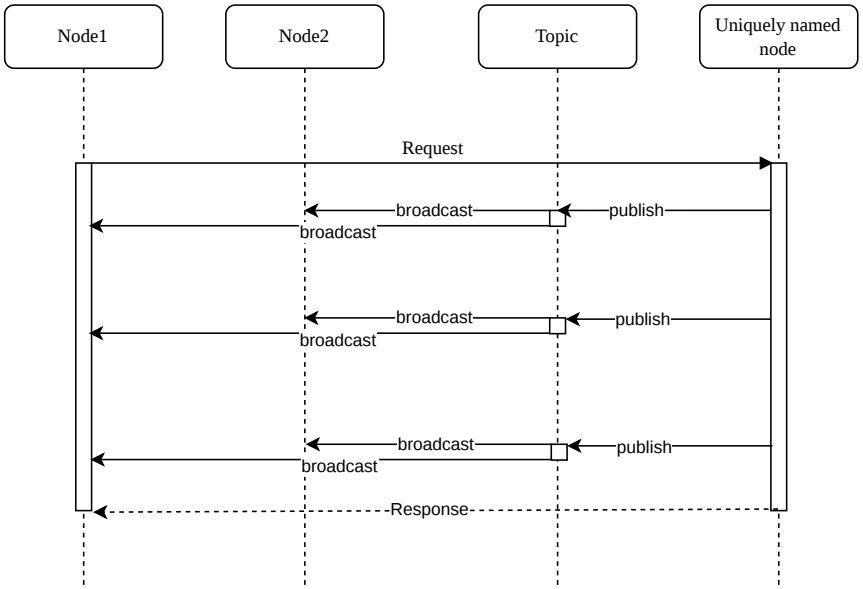


Figure 2.8: ROS actions protocol

initially developed for POSIX based operating systems, some clock and timing functions remain of the essence. This means that the microcontroller has to either be able to run a RTOS providing these functions or have abstractions providing them. Currently micro-ROS supports some Real-Time Operating Systems, such as FreeRTOS, Zephyr and NuttX, as well as some boards such as Espressif ESP32, Raspberry Pi Pico, some Arduino boards and some STM boards. There is also experimental support for bare-metal usage, other RTOSes, and other boards but these often have to use a bridge providing the necessary abstractions, and it is not always guaranteed to work due to hardware limitations.

Whilst micro-ROS tries to provide as many of the features that ROS 2 provides, there are some considerations that have had to be made, seeing as the environment has resource constraints.

- The messages interchanged between two nodes, including at least one micro-ROS node must be a fixed size. This is to avoid dynamic memory allocations, which is heavily discouraged to do on a microcontroller post build time. This is because it is a best practice to allocate all the memory you need upon initialization, as dynamic memory allocations during runtime could yield undefined behaviour when one runs out of memory to allocate.
- To deal with different mechanisms and scheduling APIs of the underlying RTOS, rclc provides a flexible Executor on the level of individual callbacks.
- ROS 2 provides a launch system to start up the nodes of the ROS node graph. Micro-ROS does not have this due to different underlying RTOS mechanisms and leaves it up to the developer.
- Standard ROS 2 allows for dynamic loading of nodes into running processes. Since the

Category	ROS 1	ROS 2
Network transport	Bespoke protocol built on TCP/UDP	Existing standard (DDS), with abstraction supporting addition of others
Network architecture	Central name server (roscore)	Peer-to-peer discovery
Platform support	Linux	Linux, Windows, and macOS
Client libraries	Written independently in each language	Sharing a common underlying C library (rcl)
Node versus process	Single node per process	Multiple nodes per process
Threading model	Callback queues and handlers	Swappable executor
Node state management	None	Lifecycle nodes
Embedded systems	Minimal experimental support (rosserial)	Commercially supported implementation (micro-ROS)
Parameter access	Auxiliary protocol built on XMLRPC	Implemented using service calls
Parameter types	Type inferred when assigned	Type declared and enforced

Table 2.2: Summary of ROS 2 features compared with ROS 1 [26].

program has to be flashed onto the microcontroller, the node composition of the micro-ROS program has to be known ahead of time, and thus at compile-time.

- Due to hardware limitations, most microcontrollers are unable to handle cryptographic libraries, leading to DDS-XRCE not being secure by default and instead delegating this responsibility to the underlying transport medium. Since a microcontroller running micro-ROS talks to one micro-ROS agent on a PC, the PC running ROS 2 can then encrypt the message since it uses DDS-security. It leaves to check if the underlying transport medium from microcontroller to PC is safe. Since eavesdropping wireless transport is trivial, all wireless transport mediums (eg: Bluetooth) encrypt packets over the air. Wired transport mediums are inherently safe, as they are physically contained, though some mediums do run some type of encryption.

To summarize, ROS is an SDK which provides modules to aid robotics development. Most notably is their communication layer which allows for peer-to-peer communication. ROS also provides an SDK for development on microcontrollers, though not all microcontrollers. Since we can leverage its communication layer to form an RP dependency graph-like structure, we could in theory convert an RP program into a ROS program, as well as send parts of the RP program to microcontrollers, using the micro-ROS SDK.

2.3.3 YARP

Let us talk about Yet Another Robotics Platform or YARP. Created for and by researchers in humanoid robots, YARP is yet another robotics platform aimed for long-term software development. YARP was created for applications that are real-time, computation-intensive and

involve interfacing with diverse and changing hardware [6]. The robotics platform was designed with the following things in mind:

- A robot control system consists of multiple processes. This means that the developer must be able to design the system as a set of processes running on a set of computers. This maximizes the amount of time spent researching instead of optimizing code. Consequently, writing and running processes has to be as easy as possible.
- Whenever we have a single process with a single responsibility, we can reuse these processes and thus have a modular process, which can be used multiple times within the same system.
- Since the actual performance of a robot controller depends on the timing of various signals, existing processes should feel minimal interference when another process is created within the system.
- Since some processes could depend on other processes, it would be ideal to minimize these dependencies to prevent expensive chains or process restarts.
- Supporting libraries should be able to be used cross-platform and external libraries should be able to be used within a YARP system, independent of YARP.
- YARP should be OS independent. Sometimes some tools are better in a one OS rather than another. YARP uses an open source library that provides a concurrent framework across a wide variety of operating systems [6].

YARP processes communicate with each other by making use of the Observer pattern and port objects. Port objects, not to be confused with actual ports on a computer, are active objects that manage multiple connections. Each connection has a state that can be manipulated from external commands, which manage the connection or obtain state information from it. Ports can either be declared as an input, or as an output. Both input and output ports can receive and send, from and to different protocols respectively. YARP supports the following protocols:

- TCP, the reliable protocol, often used in web browsers. TCP guarantees delivery.
- UDP, a faster protocol than TCP but does not have delivery guarantee.
- Multicast, used for one-to-many connections. This is efficient when wanting to share the same information across many processes.
- Shared memory. This is used for local connections when possible.
- QNet, A fast and synchronous protocol.

YARP is flexible in the sense that it allows the programmer to either declare the port connections programmatically or at runtime. By default, ports are made to process updates and send updates often, as it is more important to be up-to-date with the information, than to process all the incoming or outgoing information. This behaviour can be altered but will introduce an overhead, similar as to UDP vs TCP.

The underlying transport layer of ports are sockets, where an initial handshake is done to agree on the exchanged datatype as well as registration. As YARP is provided in C++, all non-pointer datatypes can easily be transmitted through ports. In case that the developer does have pointer based datatypes that must be transmitted, then the developer is responsible for providing serialization and deserialization functions.

Since YARP is primarily focused on humanoid robots, it provides libraries for image processing and device drivers, as missing these are unthinkable within the humanoid domain. Unfortunately, like OROCOS, YARP does not support microcontrollers thus disqualifying this robotics platform for this thesis. Like OROCOS, it may prove to be useful if the domain of RP would extend to the domain where YARP is used.

2.4 Revisiting the Problem Statement

2.5 Tierless Programming

Right now we have compared some RP languages, more specifically, some that run on microcontrollers. We have compared some robotics platforms, whose goal is to aid in robotics development. However, robotics systems are often distributed, some parts are responsible for sensing the environment, some are responsible for reacting to it, with others responsible for everything in between. This would mean that we have to create different RP programs, and somehow connect them as they are deployed. This seems rather inefficient, and to elaborate further we would like to introduce the concept of tierless programming or multitier programming [30].

The goal is simple. We have different parts of a program that live separate on different computers yet are coupled by a network. In traditional tierless systems, such as a web environment, a developer would have to not only develop client side code as well server side code, but also code glueing these tiers together. This gives the illusion that these tiers are decoupled from each other while in actuality they work closely together and actually also depend on each other. To avoid writing extra glue code or huge API boilerplates we could instead unify these tiers into a single source language.

Writing a tierless program in a single source language not only saves the developer from writing a lot of boilerplate code or glue code, it also abstracts away the low-level details relevant to the distributed program. This allows the developer to focus purely on the high level interactions and not worry about error-prone aspects like network communication. Additionally, it saves the developer from having to design inter-tier APIs, as the program is already inherently intertwined in the source file. How the tiers are connected becomes a compilation step rather than a development step.

In traditional distributed programming, the boundaries between hosts or functionalities may not always coincide. Meaning a single functionality may be distributed across many places and many functionalities could be isolated to a single place. Whenever functionalities are split across different devices, changing them could become cumbersome as the same functionality is

implemented multiple times. Using tierless programming would allow the developer to implement it once and distribute it where it is needed. Furthermore, an update to the said functionality would also update all the functionalities that are distributed in the system.

Reasoning about a distributed system is complex. You could visualize a distributed system as a graph, where the edges between the nodes would show inter-tier communication. These edges could show that the system has a series of one-to-one, one-to-many or many-to-many relationships. Reasoning about all these different codebases or systems could become cumbersome as the edges increase and thus also the complexity. With tierless programming, you would not have to context switch between these systems or codebases as the entire system or components that compose the system are declared in a single source. The same goes for code comprehension. It is a lot easier when everything is grouped together instead of scattered.

Tierless programming languages can be categorized under the following categories:

- Degrees of tierless programming
- Placement strategy
- Placement specification & granularity
- Communication abstractions
- Supported topologies.

Degree of multitier programming

Tierless programming languages either provide support for explicit placement of distribution by means of for example an annotation such as `@server` or `@client`, or they split code automatically by means of static analysis. Alternatively, some tierless languages translate code across target platforms (eg: Java to JavaScript).

Placement strategy

The placement strategy is the strategy adopted by the tierless language to split the program up. A Tierless program can be partitioned into different programs (statically or dynamically). Another option would be to separate the program into different stages, where the execution of one stage generates the next stage and can inject values computed in the current stage into the next.

Placement specification & Granularity

The tierless languages that allow for explicit placement, can optionally also allow varying degrees in granularity of the placement. Some tierless languages could for example only allow complete blocks to be placed together on a tier whilst other languages allow local bindings to be placed on different tiers.

Communication abstractions

Tierless languages also differ on the underlying communication used, so could one language use Remote Procedure Calls (RPC), asynchronous callbacks, directional message-passing channels, pub-sub mechanisms, distributed shared state or even FRP streams.

Supported topologies

Another categorizing property of tierless programming is their supported topologies. Most web-oriented tierless languages focus on the client-server architecture. This means that most of the time they're exclusively meant for that architecture, and sometimes also allow integration of a database. Some tierless languages allow specification of tiers, such as ScalaLoci [9].

```
@multitier object Chat {
    @peer type Server <: { type Tie <: Multiple[Client] }
    @peer type Client <: { type Tie <: Single[Server] }

    val message = on[Client] { Evt[String] }

    val publicMessage = on[Server] {
        message.asLocalFromAllSeq map { case (_, message) => message }
    }

    def main() on[Client] {
        publicMessage.asLocal observe println
        for (line ← io.Source.stdin.getLines)
           message.fire(line)
    }
}
```

Listing 2.10: Tierless program example from the ScalaLoci tutorial

To give an example how such a tierless program is written, consider the following example in ScalaLoci defined in Listing 2.10. It consists of a server and multiple clients where message events are sent to the server, who then makes clients print it to their consoles. By using `@peer` annotations we can define the structure of the tierless program. We can define whether there is multiple of a tier or if there is a single one, like multiple clients and a single server in this case. The cross-tier boundaries are generated by the compiler as needed.

Chapter 3

μ-Remus

3.1 Haai & Remus

3.2 ROS & μ-ROS

3.3 Mapping RP to ROS & μ-ROS

3.4 μ-Remus: A C implementation

Chapter 4

Programming Robots with Reactors

4.1 Linking ROS and μ-ROS

Chapter 5

Tierless Haai

5.1 Haai to CHaai

Chapter 6

Evaluation

6.1 Evaluating Haai distribution

6.2 Evaluating Haai to CHaai

6.3 Evaluating CHaai on μ-Remus

Chapter 7

Conclusion & Future Work

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